# A Two-Time-Level Semi-Lagrangian Semi-implicit Scheme for Spectral Models

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ABSTRACT

Recently, it has been demonstrated that the semi-implicit semi-Lagrangian technique can be successfully coupled with a *three*-time-level spectral discretization of the barotropic shallow-water equations. This permits the use of time steps that are much larger than those permitted by the Courant-Friedrichs-Lewy (CFL) stability criterion for the corresponding Eulerian model, without loss of accuracy.

In this paper we show that it is possible to further quadruple the efficiency of semi-implicit semi-Lagrangian spectral models beyond that already demonstrated. A doubling of efficiency accrues from the use of the stable and accurate *two*-time-level scheme described herein. For semi-implicit semi-Lagrangian spectral models a further doubling of efficiency can be achieved by using a smaller computational Gaussian grid than the usual one, without incurring the significant loss of stability and accuracy that is observed for the corresponding Eulerian spectral model in analogous circumstances.

## 1. Introduction

The need for higher resolution numerical weather forecasts motivates the development of more efficient time integration schemes. A significant step in this direction was the introduction of the semi-implicit scheme (Robert 1969; Kwizak and Robert 1971; Robert et al. 1972) for primitive equation models. This scheme removes the stability constraint imposed by the fast-moving gravitational oscillations by effectively slowing them down. Stability is then limited by the Courant-Friedrichs-Lewy (CFL) stability criterion associated with horizontal advection. The time truncation errors are, nevertheless, still much smaller than the spatial truncation errors.

This latter observation motivated Robert (1981, 1982) to propose coupling a semi-Lagrangian treatment of advection (where stability is no longer limited by the magnitude of the horizontal wind) with a semi-implicit treatment of gravitational oscillations. This idea was first demonstrated in the context of a shallow-water equations grid point model, and later extended to the baroclinic primitive equations (Bates and McDonald 1982; Robert et al. 1985).

During the last several years, a number of investigators have pursued the stability advantages of semi-Lagrangian advection in different contexts. Further applications of this technique to grid point models may be found in Bates (1984), McDonald (1986), Ritchie (1986) and Bates (1988). Recently the method has been extended to include both finite-element (Staniforth and Temperton 1986) and spectral (Ritchie 1987, 1988) methodologies. Semi-Lagrangian advection has also been proposed for pollutant transport problems (Pudykiewicz et al. 1985) and moisture advection (Ritchie 1985).

Present semi-Lagrangian schemes are based on discretization over either two or three time levels. Two-time-level schemes are potentially twice as efficient and simpler to code than three-time-level schemes and additionally have no time computational modes. In practice however, the advantages of two-time-level schemes over three-time-level schemes were not fully realized by the early two-time-level schemes, due to the difficulty in obtaining sufficiently accurate ($O(\Delta t^2)$) trajectories (Staniforth and Pudykiewicz 1985; McDonald 1987). Temperton and Staniforth (1987) and McDonald and Bates (1987) independently proposed an $O(\Delta t^2)$ accurate method of obtaining the trajectories for two-time-level models, and this approach has been pursued by Purser and Leslie (1988) for grid point regional models. Such an approach offers a potential doubling of efficiency with respect to three-time-level schemes, as was demonstrated by Temperton and Staniforth (1987) for a shallow-water finite-element model.

In the two-time-level scheme of Temperton and Staniforth (1987), all the linear terms are treated implicitly, while the nonlinear terms are either treated explicitly using time-extrapolated values or implicitly via iteration. The resulting algorithm is unconditionally stable (in the linear sense), which allows the time step

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to be chosen on the basis of accuracy alone. This work extends this approach to a global spectral model of the shallow-water equations.

## 2. Model formulation

### *a. Governing equations*

The governing equations are the shallow-water equations on a rotating sphere of radius $a$. We write the momentum equation in three-dimensional vector form using the undetermined Lagrange multiplier method as in Côté (1988), and the continuity equation in logarithmic form as in Temperton and Staniforth (1987). Thus, in standard notation

$$\frac{d}{dt}\mathbf{V} = \mathbf{F} + \mu\mathbf{r}, \tag{1}$$

$$\frac{d}{dt}\ln\phi + \nabla\cdot\mathbf{V} = 0. \tag{2}$$

In the above, $\mathbf{r}$ is the normalized position vector whose components in a Cartesian frame of reference fixed at the center of the sphere are,

$$\begin{aligned} x &= \cos\lambda\,\cos\theta, \\ y &= \sin\lambda\,\cos\theta, \\ z &= \sin\theta, \end{aligned} \tag{3}$$

where $\lambda$ is the longitude and $\theta$ the latitude. Also

$$\mathbf{V} = a\frac{d}{dt}\mathbf{r}, \tag{4}$$

$$\mathbf{F} = -2\Omega\,\sin\theta\mathbf{r}\times\mathbf{V} - \nabla\phi, \tag{5}$$

and $\mu$ is a Lagrange multiplier determined by requiring that the fluid elements remain on the sphere, i.e.,

$$\mathbf{r}\cdot\mathbf{r} = 1, \tag{6}$$

for all time. The Lagrange multiplier could be determined at this stage, and this leads to the usual Eulerian form for the governing equations (Côté 1988). However, as noted in Ritchie (1988), the evaluation of the nonlinear metric terms of the momentum equations at the midpoint of the trajectory leads to an instability. Ritchie (1988) described one way of resolving this problem, viz., by introducing moving tangential plane coordinates. In the present work we adopt the alternative solution proposed by Côté (1988), viz., to time discretize *before* determining the Lagrange multiplier.

The constraint (6) implies that $\mathbf{V}$ is tangent to the sphere at $\mathbf{r}$, and therefore

$$\mathbf{r}\cdot\mathbf{V} = 0. \tag{7}$$

This means that the three-dimensional wind vector has only two independent components which we write in terms of the wind images $U$ and $V$. Thus,

$$\mathbf{V} = \frac{1}{\cos\theta}(U\hat{e}_\lambda + V\hat{e}_\theta) \tag{8}$$

where $\hat{e}_\lambda$ and $\hat{e}_\theta$ are eastward and northward pointing unit vectors, respectively. Introducing the Helmholtz decomposition of $\mathbf{V}$ in terms of the streamfunction $\psi$ and the velocity potential $\chi$, we have

$$\mathbf{V} = \mathbf{r}\times\nabla\psi + \nabla\chi, \tag{9}$$

$$U = \frac{1}{a}\left(\frac{\partial\chi}{\partial\lambda} - \cos\theta\frac{\partial\psi}{\partial\theta}\right), \tag{10a}$$

$$V = \frac{1}{a}\left(\frac{\partial\psi}{\partial\lambda} + \cos\theta\frac{\partial\chi}{\partial\theta}\right), \tag{10b}$$

$$\zeta = \mathbf{r}\cdot(\nabla\times\mathbf{V}) = \nabla^2\psi, \tag{11a}$$

$$D = \nabla\cdot\mathbf{V} = \nabla^2\chi, \tag{11b}$$

where $\nabla$ and $\nabla^2$ reduce to the usual horizontal gradient and Laplacian operators on the sphere, because of the constraint (6).

### *b. Semi-Lagrangian discretization*

If we know the trajectory of the fluid element that is at position $\mathbf{r}$ at forecast time $t$, then Eqs. (1) and (2) may be integrated following the motion over a time interval $\Delta t$. Thus

$$\mathbf{V} - \frac{1}{2}\Delta t\mathbf{F} = \mathbf{V}^- + \frac{1}{2}\Delta t\mathbf{F}^- + \frac{1}{2}\mu\Delta t(\mathbf{r} + \mathbf{r}^-) \equiv \mathbf{R}, \tag{12}$$

$$\ln\phi + \frac{1}{2}\Delta tD = \ln\phi^- - \frac{1}{2}\Delta tD^-, \tag{13}$$

where $\mathbf{r}^-$ is the upstream position at time $t - \Delta t$ of the fluid element arriving at $\mathbf{r}$ at time $t$, and the negative superscript denotes evaluation at position $\mathbf{r}^-$ and time $(t - \Delta t)$. Interpolation at upstream points is bicubic as in Ritchie (1988). The Lagrange multiplier is determined by projecting the discretized equation (12) onto $\mathbf{r}$. Since for constrained motion $\mathbf{V}$ and $\mathbf{F}$ are both orthogonal to $\mathbf{r}$, then so is $\mathbf{R}$ and

$$\mu = -\frac{2}{\Delta t}\frac{\mathbf{r}\cdot\left(\mathbf{V}^- + \frac{1}{2}\Delta tF^-\right)}{(1 + \mathbf{r}\cdot\mathbf{r}^-)}. \tag{14}$$

### *c. Trajectory calculations*

To complete the discretization we need to find the trajectory. A sufficiently accurate estimate of the trajectory is obtained by integrating (1) and (4) subject to (6) over the interval $[t - \Delta t, t]$ while neglecting $\mathbf{F}$. The solutions are great circles whose parameters are determined by the wind field at $(t - \frac{1}{2}\Delta t)$ for a centered scheme. The midpoint $\mathbf{r}^0$ of the trajectory is obtained by iteratively solving

$$\mathbf{r}^0 = \cos\Theta\mathbf{r} + \sin\Theta\mathbf{r} \times \frac{(\mathbf{r} \times \mathbf{V}^0)}{|\mathbf{r} \times \mathbf{V}^0|}, \tag{15}$$

where

$$\mathbf{V}^0 = \mathbf{V}\left(\mathbf{r}^0, t - \frac{1}{2}\Delta t\right), \tag{16}$$

$$\Theta = \frac{|\mathbf{V}^0|}{a}\frac{\Delta t}{2}. \tag{17}$$

Geometrically, we search for a point $\mathbf{r}^0$ situated a distance $\frac{1}{2}|\mathbf{V}^0|\Delta t$ upstream on the sphere, such that $\mathbf{r}$, $\mathbf{r}^0$ and $\mathbf{V}^0$ are coplanar (see Fig. 1). The point $\mathbf{r}^-$ is then

$$\mathbf{r}^- = 2(\mathbf{r}\cdot\mathbf{r}^0)\mathbf{r}^0 - \mathbf{r}. \tag{18}$$

Equation (15) differs from Ritchie (1987), where the trajectories are first approximated by straight lines on a tangent plane followed by a correction to come back to the sphere; the computational cost of the two algorithms is almost identical. The impact of algorithmic choice is also negligible, giving rise to rms height differences of only 20 cm for a 5-day integration of a global T126 spectral model with a 2 h time step.

As in Temperton and Staniforth (1987) we use time-extrapolated values for $\mathbf{V}^0$. This is the key step of the algorithm that allows a saving of a factor of 2 over the usual semi-implicit semi-Lagrangian algorithm where we need to forecast the wind at time $(t - \frac{1}{2}\Delta t)$. We examine constant (one-term), linear (two-term) and quadratic (three-term) extrapolation, viz.

$$\mathbf{V}\left(\mathbf{r}, t - \frac{1}{2}\Delta t\right) = c_1\mathbf{V}(\mathbf{r}, t - \Delta t) + c_2\mathbf{V}(\mathbf{r}, t - 2\Delta t) + (1 - c_1 - c_2)\mathbf{V}(\mathbf{r}, t - 3\Delta t), \tag{19}$$

with $(c_1, c_2) = (1, 0)$, $(3/2, -1/2)$ and $(15/8, -10/8)$, respectively.

This step involves the wind at previous time steps and the scheme is no longer formally a two-time-level scheme. However, once the trajectories have been determined, the momentum and continuity equations are discretized as time differences and averages over just two time levels ($t$ and $t - \Delta t$), and we subsequently refer to the scheme as being two-time-level. In effect, we replace the full-cost odd time steps of the three-time-level scheme by an almost-free time extrapolation of the winds.

The convergence of the trajectory calculation was investigated in Pudykiewicz et al. (1985), in the context of a Cartesian polar-stereographic-grid model; an upper bound of 6 hours for the time step ($\Delta t$) was deduced as a sufficient condition for convergence to a unique solution. However, even with such a time step, the metric effects should be small, and this estimate may also be expected to apply to the iterative scheme (15), especially when using values at the previous time step as an educated first guess.

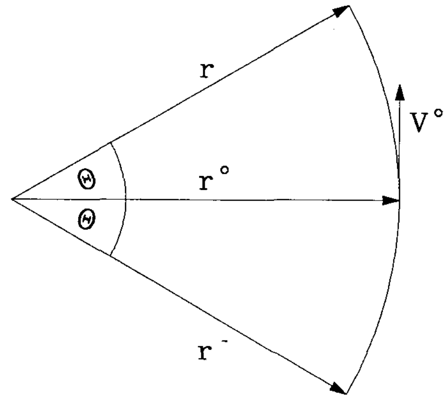


FIG. 1. Schematic diagram for the geometry of the trajectory calculations.

*d. Spectral solution*

We first recast Eqs. (12) and (13) in a form more suitable for solution by the spectral method. We introduce a reference geopotential $\phi^*$ and define $\phi'$ as the departure of $\phi$ from $\phi^*$. Operating on (12) with $\mathbf{r}\cdot(\nabla\times)$ and $\nabla\cdot$ successively, multiplying both sides of (13) by $\phi^*$ and transferring the nonlinear terms to the right-hand side, we finally obtain

$$\zeta + \alpha\sin\theta D + \alpha\frac{V}{a} = L, \tag{20}$$

$$D - \alpha\sin\theta\zeta + \alpha\frac{U}{a} + \beta\nabla^2\phi' = M, \tag{21}$$

$$\phi' + \gamma D = Q, \tag{22}$$

where

$$\alpha = \Omega\Delta t, \quad \beta = \frac{1}{2}\Delta t, \quad \gamma = \frac{1}{2}\phi^*\Delta t, \tag{23}$$

and

$$L = \mathbf{r}\cdot(\nabla \times \mathbf{R}), \tag{24}$$

$$M = \nabla\cdot\mathbf{R}, \tag{25}$$

$$Q = \left[\phi' - \phi^*\ln\left(1 + \frac{\phi'}{\phi^*}\right)\right] + \left[\phi^*\ln\left(1 + \frac{\phi'}{\phi^*}\right) - \gamma D\right]^-. \tag{26}$$

Equations (20)–(26) differ in two aspects from the usual spectral discretization of the shallow-water equations. First, by the presence of the Coriolis terms ($\alpha$) on the left-hand side of (20) and (21) and, second, by the nonlinear dependence of the right-hand side of (22) on the solution $\phi'$ (first square-bracketed term).

The first point is resolved by a generalization of the Helmholtz solver to include the Coriolis terms. The method is outlined in the Appendix. It is equivalent to a spectral model that uses truncated Hough functions as in Kasahara (1977).

The second point is handled by iteratively solving (20)–(22) for $\phi'$ with an appropriate initial guess for $Q$. We use

$$Q_{\text{initial guess}} = (\phi' - \gamma D)^{-} - \Delta t(\phi' D)^{0}, \quad (27)$$

the usual semi-implicit expression, but with $\phi'^{0}$ and $D^{0}$ obtained via time extrapolation of mesh values of $\phi'$ and $\mathbf{V}$ [c.f. (19)] followed by interpolation to the midpoint ($\mathbf{r}^{0}$) of the trajectory.

The rest of the spectral discretization is similar to that described in Ritchie (1988). It is a triangularly truncated global spectral model with a computational Gaussian grid.

### *e. Reduced-resolution Gaussian grid*

For Eulerian advection in a spectral model, the minimum resolution of the computational Gaussian grid required to prevent aliasing (and computational instability) is determined by the order of the product terms (Orszag 1970; Machenhauer 1979). Because the advection terms of the shallow-water equations are quadratic, it is necessary to use a computational Gaussian grid having at least $(3/2)^2$ more points than the minimum-size grid required for transformations of a field from spectral to grid point space and back again without loss of information.

However, for semi-Lagrangian advection in general (and in the context of a spectral model in particular) the nonlinearity of advection is "hidden" in the interpolations. Since these interpolations implicitly contain some damping (Bates and McDonald 1982; McDonald 1984; Pudykiewicz and Staniforth 1984), and since semi-Lagrangian finite-difference and finite-element schemes have been found to be computationally stable, it seems plausible that a reduced-resolution Gaussian grid would be sufficient for the treatment of the advection terms of a semi-Lagrangian spectral model. The minimum sized grid would be one that is required to transform from spectral to grid point space and back again without loss of information. It would have approximately one-half [more precisely $(2/3)^2$] the number of points as the grid usually chosen, and would represent exactly all linear terms. Further, an examination of the governing equations after semi-Lagrangian discretization [Eqs. (20)–(26)] shows them to be linear apart from the logarithmic terms in (26). These terms are linear to order $(\phi'/\phi^*)^2$ and are consequently very weakly nonlinear. Thus, it also seems plausible that the remaining terms would not suffer unduly from such a reduced-resolution choice for the computational Gaussian grid and a further halving of the computational cost of the model would ensue.

The semi-Lagrangian interpolations would, of course, be somewhat less accurate due to the use of a reduced-resolution computational grid, but this effect is probably unimportant since it should have negligible impact on all scales, except the very smallest (which have little predictive value in any case). The idea of using a reduced-resolution Gaussian grid is explored experimentally in section 4.

## 3. Analysis

A model time step may be summarized as

$$X = \left[1 + \tfrac{1}{2} i\Delta t H\right]^{-1} (Y + N), \quad (28)$$

where

$$X = \begin{pmatrix} \zeta \\ D \\ \phi' \end{pmatrix}, \text{ unknown fields at time } t$$

$Y$ = known fields evaluated using quantities at preceding time steps $(t - \Delta t)$, $(t - 2\Delta t)$, etc

$$N = \begin{bmatrix} 0 \\ 0 \\ \phi' - \phi^* \ln\left(1 + \dfrac{\phi'}{\phi^*}\right) \end{bmatrix}$$

and $H$ is a constant-coefficient linear operator. The eigenfunctions (normal modes) of $H$ are Hough functions whose corresponding eigenvalues ($\omega$) are real for positive $\phi^*$.

### *a. Linear stability*

We examine the linear stability of the time stepping algorithm by linearizing about a resting basic state of constant geopotential $\phi^*$. On the right-hand side of (28), $N$ drops out and $Y$ reduces to

$$Y = \left[1 - \tfrac{1}{2} i\Delta t H\right] X(t - \Delta t). \quad (29)$$

The normal modes are found by writing

$$X(t) = EX(t - \Delta t),$$
$$E = e^{-iW\Delta t}, \quad (30)$$

where $E$ is the amplification factor and $W$ the frequency of the normal mode. It is easily verified that Hough functions are also the normal modes of the discretized linear equations and the amplification factor is related to the frequency $\omega$ by

$$E = \left(\frac{1 - \frac{1}{2} i\omega\Delta t}{1 + \frac{1}{2} i\omega\Delta t}\right), \quad (31)$$

and therefore

$$W = \frac{2}{\Delta t} \tan^{-1}\left(\frac{1}{2} \omega\Delta t\right) = \omega - \omega^3 \frac{\Delta t^2}{12} + O(\Delta t^4). \quad (32)$$

The amplification factor being of unit modulus, the scheme is neutrally stable and $O(\Delta t^2)$ accurate. This analysis, because of the assumed resting basic state, does not take into account the damping due to the interpolation of semi-Lagrangian advection when the departure points are no longer mesh points. This aspect of the problem is discussed in detail in McDonald (1984) and Ritchie (1986), and their analyses are also applicable to the present work.

*b. Convergence of the iterations*

At each time step, (28) is solved iteratively by successive approximation, starting from an initial guess, i.e.,

$$X^{(n)} = R(X^{(n-1)}), \quad n = 1, 2 \cdots. \tag{33}$$

We now examine the convergence of this iterative process. Making a variation of the right-hand side, we obtain to first order,

$$\delta X^{(n)} = \left[1 + \frac{1}{2} i\Delta t H\right]^{-1} \left(\frac{\phi'}{\phi}\right)^{(n-1)} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \delta X^{(n-1)}. \tag{34}$$

A sufficient condition for convergence to a unique solution independent of the initial guess, is that the operator acting on $\delta X^{(n-1)}$ be a damping operator. This will certainly be the case if

$$\rho = \max\left(\left|\frac{\phi'}{\phi}\right|\right) < 1, \tag{35}$$

since the spectral radius of $[1 + \frac{1}{2} i\Delta t H]^{-1}$ is

$$\max\left|\frac{1}{1 + \frac{1}{2} i\omega\Delta t}\right| = \max \frac{1}{\left(1 + \omega^2 \frac{\Delta t^2}{4}\right)^{1/2}} \leqslant 1. \tag{36}$$

It is easy to find a value of $\phi^*$ such that (35) is satisfied. It is preferable, however, to make a choice that will maximize the rate of convergence by minimizing $\rho$. Assuming that $\phi$ varies between $\phi_{\min}$ and $\phi_{\max}$ then

$$\rho_{\min} = \frac{\phi_{\max} - \phi_{\min}}{\phi_{\max} + \phi_{\min}}, \tag{37}$$

$$\phi^* = \frac{1}{2}(\phi_{\min} + \phi_{\max})(1 - \rho_{\min}^2). \tag{38}$$

If the variation of $\phi$ is small, with respect to the mean, then $\rho_{\min}^2$ is much smaller than 1 and, to a good approximation, the optimum $\phi^*$ is the minimax value $\frac{1}{2}(\phi_{\min} + \phi_{\max})$. This is consistent with the stability constraint of the usual semi-implicit algorithm (Simmons et al. 1978; Côté et al. 1983), viz.,

$$\phi^* \pm \phi' \geqslant 0. \tag{39}$$

*c. Bilinear formulation of the continuity equation*

An alternative to the iterative solution of (28) is to simply use the explicit bilinear formulation given by (22) and (27) without iteration. The bilinear term, however, strongly affects the gravity modes and the impact of such a procedure might prove dangerous. We study this aspect by performing a linear stability analysis about a resting basic state of constant geopotential $\phi^* + \phi_e$, and since the Coriolis terms are of secondary importance in this problem, we set them to zero. The normal modes of the linearized equations are then the spherical harmonics ($Y_l^m$). The Rossby or slow modes are stationary and therefore stable. The amplification factors of the gravity modes are the solutions of

$$(E-1)^2 + x(E+1)^2 + 2y(E+1)\left[c_1 + \frac{c_2}{E} + \frac{(1-c_1-c_2)}{E^2}\right] = 0, \tag{40}$$

where

$$(x, y) = \frac{l(l+1)\Delta t^2}{4a^2}(\phi^*, \phi_e);$$

$c_1$ and $c_2$ are the extrapolation coefficients of (19), and we assume $\phi^* > 0$.

We find that two (three)-term extrapolation introduces 1(2) spurious or computational mode(s). Furthermore, in the case of one-term and two-term extrapolation, the region of stability is between the lines $y = 0$ and $y = -x$ in the $x$, $y$ plane. In this region, the usual eastward and westward propagating gravity modes are damped. For three-term extrapolation we have the same behavior except that the stability domain is delimited by the lines $y = 0$ and $y = (20 - 28x)/63$.

In summary, for the bilinear formulation, the choice $\phi^* \geqslant \phi_{\max}$ gives a stable scheme for one-term and two-term extrapolation with a damping of the gravity modes of all scales. This choice for three-term extrapolation gives damping for the small scale gravity modes and a slight amplification for the large scale ones. Thus, for three-time-level extrapolation it is not possible to choose $\phi^*$ such that all modes are stable. These conclusions have been verified experimentally.

## 4. Results

*a. Qualitative comparison of two-time level vs three-time-level forecasts*

To assess the accuracy and efficiency of the proposed two-time-level semi-Lagrangian spectral model with respect to the three-time-level semi-Lagrangian model of Ritchie (1988), we performed 5-day global integrations of the two models at a resolution of T126, starting from the same data. In the two-time-level integrations, we used two-term extrapolation to obtain trajectory winds. All integrations were verified against a T213 Eulerian control integration using a time step of 6 minutes.

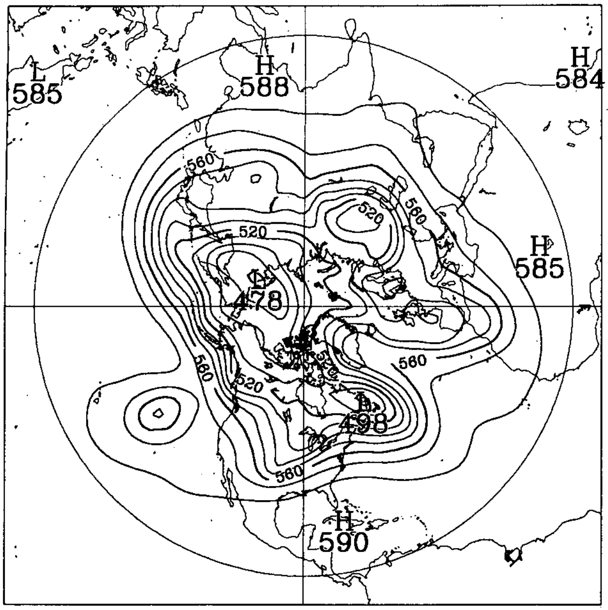


FIG. 2. Initial geopotential height in dam; contour interval = 10 dam.

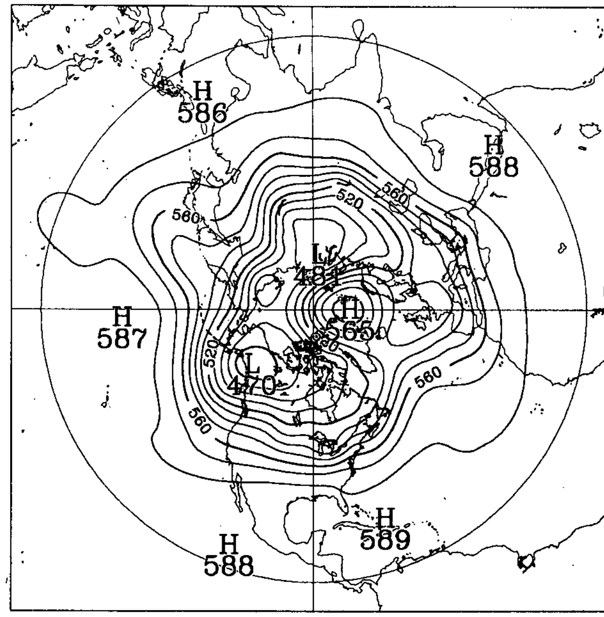


FIG. 4. As in Fig. 2, but 5-day three-time-level semi-Lagrangian T126 forecast; $\Delta t$ = 1 h.

The initial geopotential height and the resulting high-resolution Eulerian control forecast over the Northern Hemisphere are shown in Figs. 2 and 3, respectively. Figure 4 displays the three-time-level semi-Lagrangian forecast (see Fig. 5 of Ritchie 1988) performed with a time step of 1 hour, while Fig. 5 displays the two-time-level forecast performed with a time step of 2 hours. The computational cost/time step of both the three- and two-time-level schemes is almost identical, and both approximate time derivatives by time differences over the same 2-h period. However, for a forecast of given length, the two-time-level scheme requires half the number of time steps as the three-time-level scheme, and thus the two-time-level semi-Lagrangian forecast is achieved at half the computational cost of the three-time-level scheme.

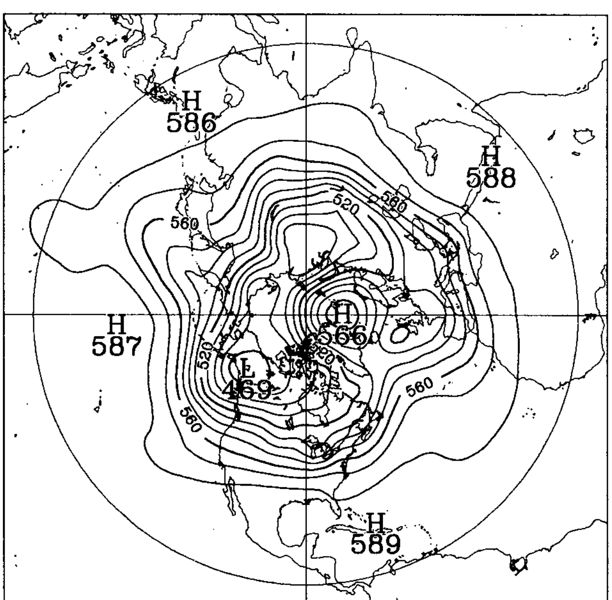


FIG. 3. As in Fig. 2, but 5-day Eulerian T213 forecast; $\Delta t$ = 6 min.

Qualitatively, one sees a great similarity between the two semi-Lagrangian forecasts, although one was made with a time step twice as large as the other. The two main features of the Eulerian control integration at 5 days are a 469 dam low over the Yukon and a 566 dam high over the Greenland Sea, which are forecast at 470 and 565 dam, respectively, by both semi-Lagrangian models. Among the differences, we note that the 490 dam contour over the Arctic Sea in the two-time-level forecast looks more like that of the control run. The 520 dam contour west of Ireland is missing from the three-time-level forecast, while it encompasses most of the United Kingdom in the two-time-level forecast. The two models have slightly different dispersion properties, as can be judged by the trough near the Great Lakes in the control run. The three-time-level model puts this feature a little ahead of the control result, while there is a slight lag in the two-time-level forecast. These differences in position are insignificant, however, when considered in the context of a 5-day forecast at this resolution.

*b. Reduced-grid forecasts*

The forecasts of the two semi-Lagrangian models displayed in Figs. 4 and 5 were both produced using a

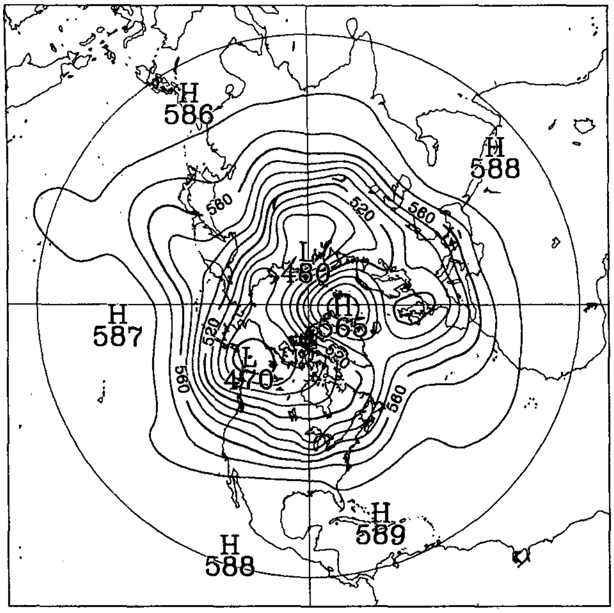


FIG. 5. As in Fig. 2, but 5-day two-time-level semi-Lagrangian T126 forecast; $\Delta t$ = 2 h.

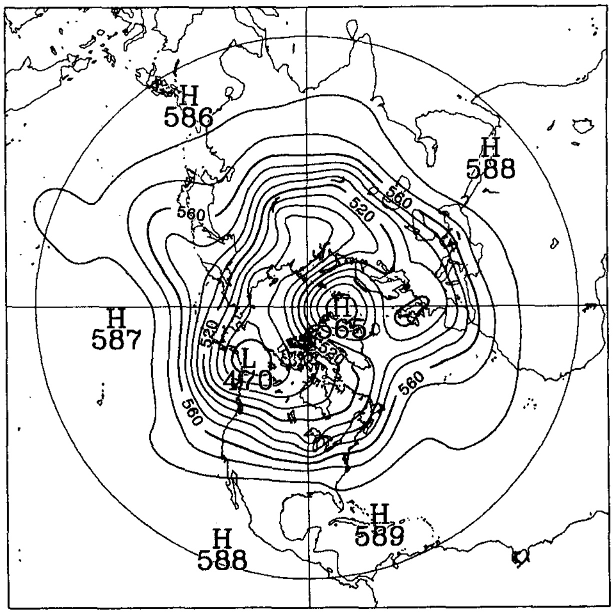


FIG. 6. As in Fig. 5, but using the reduced-size Gaussian grid.

384 × 190 computational Gaussian grid; this grid is almost identical in size to the minimum-size grid needed to prevent quadratic aliasing (and nonlinear computational instability) in an Eulerian spectral model. For semi-Lagrangian spectral models, we argued (see section 2e) the plausibility of being able to use a reduced-resolution grid without significantly affecting either accuracy or stability.

In Fig. 6 we display the 5-day forecast from the two-time-level model using a 256 × 128 reduced-resolution grid and the same time step (2 h); this grid has less than half [$\approx (2/3)^2$] the number of grid points needed by an Eulerian model at the same resolution. Comparing Fig. 6 to Figs. 4 and 5, we see that the two-time-level scheme on a reduced-resolution Gaussian grid gives a forecast of comparable quality to that obtained at twice the cost using the same scheme on the full resolution grid, and to that obtained at four times the cost by the three-time-level scheme of Ritchie (1988). We also found (not shown) that the three-time-level scheme of Ritchie (1988) also gives a 5-day forecast of comparable quality to those of Figs. 4–6 when using the reduced-resolution Gaussian grid, at a computational cost similar to that of our two-time-level scheme run on the full-resolution grid. On the other hand, the Eulerian model run on the reduced-resolution grid gives a noisy forecast at day 5 and becomes computationally unstable around day 9, whereas all of the aforementioned semi-Lagrangian models were successfully integrated to 20 days, with no signs of noise or computational instability. We would expect the noninterpolating model of Ritchie (1988), with a reduced-resolution grid, to behave like the corresponding Eulerian model, due to the aliased treatment of the perturbation advection terms.

*c. Quantitative evaluation*

For a more quantitative evaluation of the forecasts, we display in Fig. 7 the evolution of the global rms height differences between the aforementioned semi-Lagrangian runs and the control run. We see that the scores are very close indeed with the reduced-resolution three-time-level integration about 3 m less accurate by day 5 than the other three.

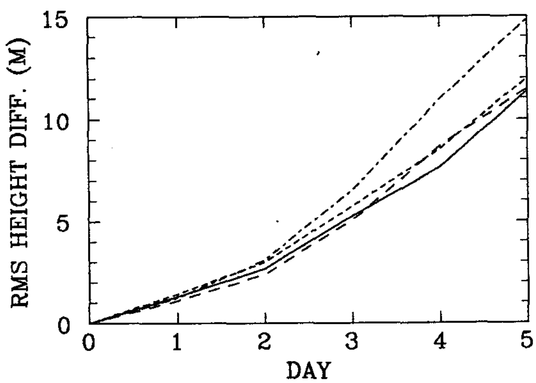


FIG. 7. Root-mean-square height differences (m) as a function of forecast duration (day) with respect to the T213 Eulerian control. Three-time-level semi-Lagrangian model, $\Delta t$ = 1 h (long dashed); two-time-level semi-Lagrangian model, $\Delta t$ = 2 h (solid); three-time-level semi-Lagrangian model using reduced-resolution Gaussian grid, $\Delta t$ = 1 h (long-short dashed); and two-time-level semi-Lagrangian model using reduced-resolution Gaussian grid, $\Delta t$ = 2 h (short dashed).

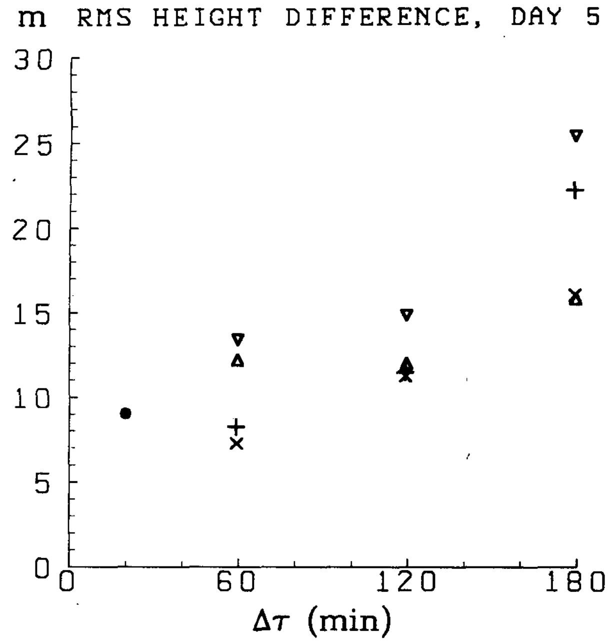


FIG. 8. Root-mean-square height differences (m) at day 5 of T126 integrations, with respect to the T213 Eulerian control, as a function of $\Delta\tau$ (min). Eulerian (·); three-time-level semi-Lagrangian (+); two-time-level semi-Lagrangian (×); three-time-level semi-Lagrangian using reduced-resolution Gaussian grid (∇); two-time-level semi-Lagrangian using reduced-resolution Gaussian grid (Δ).

In order to investigate the growth of time truncation error as a function of time step length, we ran a series of forecasts for different values of $\Delta t$ using the three-time-level scheme of Ritchie (1988) and our two-time-level scheme, on both full (384 × 190) and reduced-resolution (256 × 128) Gaussian grids. The results at 5 days are displayed in Fig. 8 where $\Delta\tau$ is the interval over which time derivatives are calculated. Thus $\Delta\tau = \Delta t$ for the two-time-level integrations but $\Delta\tau = 2\Delta t$ for the three-time-level integrations. For a given value of $\Delta\tau$, the two-time-level schemes require approximately half the work of the corresponding three-time-level schemes.

The solid circle on Fig. 8 shows the rms difference from the high-resolution (T213) control Eulerian forecast when using the same Eulerian scheme but at the resolution (T126) of the semi-Lagrangian integrations. Since the time step of the T126 Eulerian integration is necessarily small (for stability reasons) this provides an approximate measure of the spatial truncation error associated with T126 resolution.

Comparing first the two-time-level semi-Lagrangian integrations using the 384 × 190 full-resolution grid (×) against the corresponding three-time-level integration (+) using the same grid, we see that the two-time-level integrations achieve the same accuracy (for $\Delta\tau$ = 60 and 120 min), or better accuracy (for $\Delta\tau$ = 180 min), than the three-time-level integrations, but at half the computational cost.

Comparing next the corresponding two (Δ) and three (∇)-time-level integrations that use the 256 × 128 reduced-resolution Gaussian grid against one another, we find a similar behavior except that the advantage of the two-time-level scheme over the three-time-level scheme seems to be a little stronger. For small $\Delta\tau$ (60 min) the reduced-resolution integrations are a little less accurate than both full-resolution integrations. For larger $\Delta\tau$ (120 and 180 min) the two-time-level integrations suffer no degradation in quality when using the reduced-resolution grid, whereas some degradation is observed for the corresponding three-time-level schemes.

The choice of an acceptable level of accuracy (and consequently a scheme and time step) is somewhat arbitrary. For discussion purposes, we arbitrarily adopt the criterion suggested by one of our colleagues (André Robert) of 3 m/day (i.e., 15 m after 5 days) on the basis that this is insignificant when compared with typical 5-day forecast errors. Thus, for example, we see that it is possible to double the efficiency of the three-time-level scheme of Ritchie with a 1 h timestep ($\Delta t$ = 1 h, $\Delta\tau$ = 2 h), by merely using the same scheme but with the reduced-resolution Gaussian grid, and that it is possible to quadruple the efficiency by using our two-time-level scheme with a 2 h time step ($\Delta t = \Delta\tau$ = 2 h) and the reduced-resolution Gaussian grid.

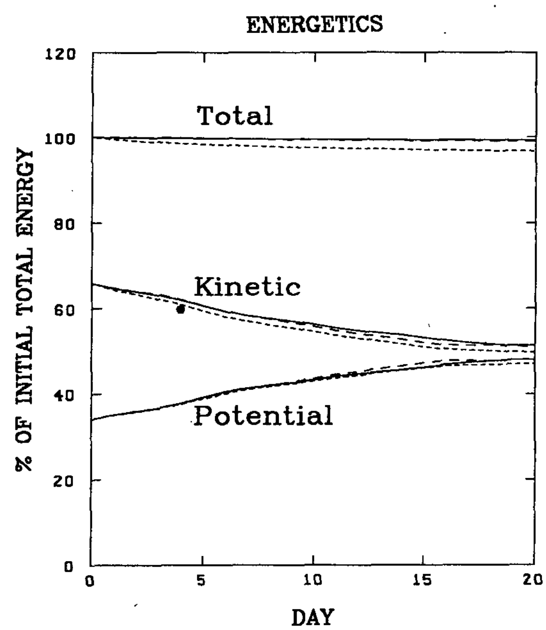


FIG. 9. Evolution of potential, kinetic and total energy in 20-day integrations, as discussed in text. Eulerian integration at T106 with explicit horizontal $\nu\nabla^4$ diffusion ($\nu = 10^{15}$ m$^4$ s$^{-1}$) and 12 min time step denoted by (short dashed); three-time-level semi-Lagrangian integration at T126 with no explicit horizontal diffusion and 1 h time step denoted by (long dashed); two-time-level semi-Lagrangian integration at T126 with no explicit horizontal diffusion and 2 h time step denoted by (solid).

### d. *Energy conservation*

Ritchie (1988) has given (his Fig. 11) the evolution of the potential, kinetic and total energy for three 20-day integrations, where the energies are expressed as a percentage of the initial total energy. The first integration, a T106 Eulerian one with a 12-min time step and a $\nu\nabla^4$ diffusion of vorticity, divergence and perturbation geopotential ($\nu = 10^{15}$ m$^4$ s$^{-1}$), was given as an indication of the energy conservation behavior that is considered to be acceptable for typical medium-range forecasts (e.g., see Jarraud et al. 1985). The other two integrations are three-time-level semi-Lagrangian schemes (interpolating and noninterpolating) at T126 with a 1 h time step. He concluded that the Eulerian control integration with diffusion loses about 3% of the total initial energy after 20 days, whereas the effect of damping due to the interpolations of the semi-Lagrangian schemes is three times smaller.

We make a similar comparison in Fig. 9 (for the 384 × 190 full-resolution two and three-time-level integrations) and 10 (for the corresponding 256 × 128 reduced-resolution schemes). We find that our two-time-level scheme with a 2 h time step conserves energy marginally better than Ritchie's three-time-level scheme. However, for both the two and three-time-level semi-Lagrangian integrations that use the reduced-resolution grid, we find that energy conservation is not as good as for the corresponding integrations that use the full-resolution grid. Nevertheless, they still conserve energy a little better than the T106 Eulerian benchmark model.

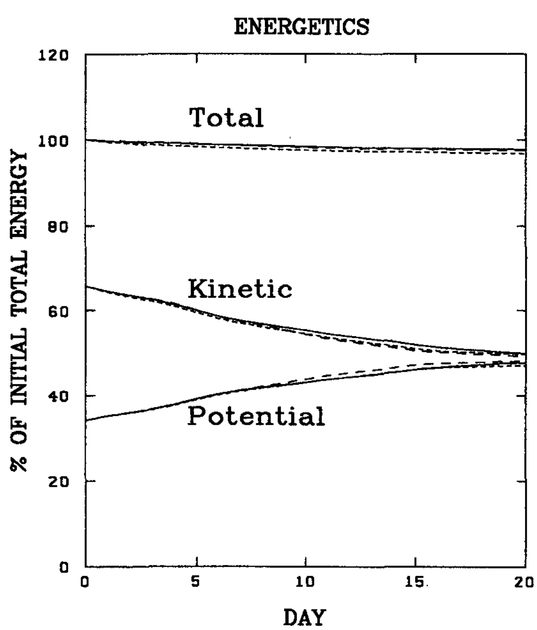


FIG. 10. As in Fig. 9 except that the two semi-Lagrangian schemes use a 256 × 128 reduced-resolution Gaussian grid instead of the 384 × 190 full-resolution one.

## 5. Concluding remarks

We have presented an efficient two-time-level scheme for barotropic spectral models that is free from any stability constraint: the choice of the time step can therefore be made on the basis of accuracy alone.

A quadrupling of efficiency is achieved with respect to the three-time-level scheme described in Ritchie (1988). A doubling of efficiency comes from the use of the two-time-level scheme described here, whereas the further doubling is due to the use of a smaller computational Gaussian grid.

The damping associated with the interpolations of the semi-Lagrangian method is less than that observed when using a horizontal diffusion parameterization typical of that used in current operational models.

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## APPENDIX

### Solution of the Generalized Helmholtz Problem

We write Eqs. (20) and (21) in terms of $\zeta$ and $D$ using (10) and (11). We then expand $\zeta$, $D$, $\phi'$, $L$, $M$ and $Q$ in spherical harmonics $[Y_l^m(\theta, \lambda)]$ and apply the Galerkin procedure to finally obtain for each value of the zonal wavenumber $m$,

$$\mathbf{A}\boldsymbol{\zeta} + \mathbf{B}\mathbf{D} = \mathbf{L} \quad \text{(A1)}$$

$$-\mathbf{B}\boldsymbol{\zeta} + \mathbf{A}\mathbf{D} + \mathbf{C}\boldsymbol{\phi}' = \mathbf{M} \quad \text{(A2)}$$

$$\gamma\mathbf{D} + \boldsymbol{\phi}' = \mathbf{Q} \quad \text{(A3)}$$

where for each field the vector notation is for the column vector of the spectral coefficients of zonal wavenumber $m$ ordered with increasing $l$, and the matrices **A**, **B** and **C** have elements

$$A(l, l') = \left[1 - \frac{im\alpha}{l(l+1)}\right]\delta_{l',l}, \quad \text{(A4)}$$

$$B(l, l') = \alpha\left[\frac{l+1}{l}\epsilon_l^{(m)}\delta_{l',l-1} + \frac{l}{l+1}\epsilon_{l+1}^{(m)}\delta_{l',l+1}\right], \quad \text{(A5)}$$

$$C(l, l') = -\frac{\beta l(l+1)}{a^2}\delta_{l',l}, \quad \text{(A6)}$$

where

$$\epsilon_l^{(m)} = \left(\frac{l^2 - m^2}{4l^2 - 1}\right)^{1/2}. \quad \text{(A7)}$$

In (A4) the square bracket evaluates to 1 when $m$ is zero.

The system (A1) to (A3) can be reduced to a linear problem involving only one unknown. Since $\mathbf{A}$ is a diagonal matrix, a convenient solution consists of expressing $\zeta$ and $\phi'$ in terms of $D$ via (A1) and (A3), viz.

$$\zeta = \mathbf{A}^{-1}(\mathbf{L} - \mathbf{BD}), \tag{A8}$$

$$\phi' = \mathbf{Q} - \gamma\mathbf{D}, \tag{A9}$$

which substituted in (A2) yield

$$[\mathbf{A} - \gamma\mathbf{C} + \mathbf{BA}^{-1}\mathbf{B}]\mathbf{D} = \mathbf{M} - \mathbf{CQ} + \mathbf{BA}^{-1}\mathbf{L}. \tag{A10}$$

The particular structure of $\mathbf{B}$ is reflected on the left-hand-side matrix of Eq. (A10), which couples only the nearest neighbors of the same $l$-parity, giving rise to two decoupled tridiagonal systems of equations. These are easily solved by direct Gauss-Jordan reduction without pivoting, owing to the diagonal dominance of the left-hand-side matrix.

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